

BI

Assignment 1:- Import Data from Different Sources in Power BI

Q1. What is Power BI?

Power BI is a Business Intelligence tool developed by Microsoft used for:

- Data analysis
- Data visualization
- Report generation
- Dashboard creation

Q2. What is legacy data?

Legacy data is old or outdated data stored in old systems or formats that are difficult to access or manage.

Q3. What are the sources of legacy data?

Sources of legacy data are:

- Relational databases
- Non-relational databases
- XML files
- Flat files
- Web services

Q4. What is OData Feed?

OData (Open Data Protocol) is a protocol used to access and exchange data over the internet.

Q5. What is the use of the “Get Data” option?

The “Get Data” option is used to connect Power BI with different data sources.

Q6. How do you import Excel data into Power BI?

Steps:

1. Open Power BI
2. Click Get Data
3. Select Excel
4. Choose Excel file
5. Select table
6. Click Load or Edit

Q7. What is the use of the “Get Data” option?

The “Get Data” option is used to connect Power BI with different data sources.

Q8. What is Power Query Editor?

Power Query Editor is used to:

- Clean data
 - Transform data
 - Filter data
- before loading it into Power BI.

Q. What is a flat file?

A flat file is a simple text file used to store data.

Examples:

- CSV file

Assignment 2:- Data Visualization from ETL Process

Q1. What is ETL?

ETL stands for:

- Extract
- Transform
- Load

It is a process used to collect, process, and store data for analysis.

Q2. What is the Extract process in ETL?

Extract is the process of collecting data from different sources such as:

- Databases
- Flat files

Q3. What is Transformation in ETL?

Transformation is the process of modifying or cleaning extracted data before loading it.

Examples:

- Removing NULL values
- Data cleansing

Q4. What is the Load process in ETL?

Load is the process of storing transformed data into the target system such as:

- Data Warehouse
- Data Mart

Q5. What is Data Warehouse?

A Data Warehouse is a centralized storage system used to store large amounts of processed data for analysis and reporting.

Q6. What is a Data Mart?

A Data Mart is a smaller section of a Data Warehouse focused on a specific department or business function.

Assignment 3:- ETL Process to Construct Database in SQL Server / Power BI

Q1. What are the types of Data Extraction?

There are two types:

1. Full Extraction
 - a. Entire data is extracted from source system.
2. Partial Extraction (Delta Load)
 - a. Only updated or changed data is extracted.

Q2. What is Data Transformation?

Data Transformation is the process of converting raw data into a proper format suitable for the target system.

It includes:

- Data cleaning
- Data mapping
- Data conversion

Q3. What is Data Loading?

Data Loading is the process of storing transformed data into the target database or data warehouse.

Q4. What is Delta Load?

Delta Load means extracting only newly updated or modified data from the source system.

Q5. What is SSDT?

SSDT (SQL Server Data Tools) is a Microsoft development environment used for:

- ETL development
- Database development
- SQL Server Integration Services (SSIS)

Q6. What is the importance of relationships in Power BI?

Relationships help connect multiple tables and allow accurate data analysis and report generation.

Q7. What is the use of “Remove Other Columns” option?

It removes unwanted columns and keeps only the selected important columns.

Q8. What is a Custom Column in Power BI?

A Custom Column is a new calculated column created using formulas.

Example:

$\text{LineTotal} = \text{UnitPrice} \times \text{Quantity}$

Assignment 4:- Data Analysis and Visualization using Advanced Excel

Q1. What is Data Visualization?

Data Visualization is the graphical representation of data using:

- Charts
- Graphs
- Maps
- Dashboards

to understand trends and patterns easily.

Q2. What is a Bar Chart?

A Bar Chart is used to compare categorical data.

- Categories are shown on Y-axis
- Values are shown on X-axis

Q3. What is a Column Chart?

A Column Chart is a vertical bar chart used to compare values across categories.

- Categories → X-axis
- Values → Y-axis

Q4. What is a Line Chart?

A Line Chart is used to show changes or trends over time.

It is useful for time-series data.

Q5. What is a Pie Chart?

A Pie Chart shows proportions or percentages of a whole.

It is used to compare contributions of categories.

Q6. What is a Scatter Plot?

A Scatter Plot is used to show the relationship between two variables and helps identify correlations.

Q7. What is a Waterfall Chart?

A Waterfall Chart shows how positive and negative values contribute to a final total.

It is commonly used in financial analysis.

Q8. What are Chart Elements in Excel?

Chart elements include:

- Chart title
- Legend
- Data labels
- Gridlines
- Axes

These improve chart readability.

Assignment 5:- Data Classification using Classification Algorithm

Q1. What is Classification?

Classification is a process of predicting the target class or category using trained data.

Example:

- Spam or Not Spam
- Male or Female
- Apple or Orange

Q2. What is a Classification Algorithm?

A Classification Algorithm is a machine learning algorithm used to assign data into predefined classes or categories.

Q3. What is a Classifier?

A Classifier is an algorithm that maps input data to a specific category or class.

Q4. What is a Feature in Classification?

A Feature is a measurable property or characteristic of data used for prediction.

Example:

- Hair length
- Color
- Weight

Q5. What is Binary Classification?

Binary Classification is a classification task with only two possible outputs.

Example:

- Yes / No
- Male / Female
- Spam / Not Spam

Q6. What is Multi-class Classification?

Multi-class Classification means classification with more than two classes.

Example:

- Cat
- Dog
- Horse

Q7. What are the applications of Classification Algorithms?

Applications:

- Email spam detection
- Sentiment analysis
- Cancer detection
- Loan prediction

Q8. Name some Classification Algorithms.

Examples:

- Logistic Regression
- Naive Bayes
- Decision Tree
- Random Forest
- K-Nearest Neighbor (KNN)
- Neural Networks

Q7. Difference between Classification and Clustering?

Classification	Clustering
Predicts predefined classes	Groups similar data
Supervised learning	Unsupervised learning
Uses labeled data	Uses unlabeled data

Assignment 6:- Data Clustering using Clustering Algorithm

Q1. What is Clustering?

Clustering is an unsupervised machine learning technique used to group similar data points into clusters based on their characteristics.

Q2. What is Unsupervised Learning?

Unsupervised Learning is a type of machine learning where data does not contain predefined labels or target outputs.

Q3. What is K-Means Clustering?

K-Means is a clustering algorithm that divides data into K groups using centroids.

Each cluster is represented by its mean point called centroid.

Q4. What is a Centroid in K-Means?

A Centroid is the center point of a cluster calculated using the mean of all data points in that cluster.

Q5. What are the steps of K-Means Algorithm?

Steps:

1. Choose number of clusters (K)
2. Initialize centroids
3. Assign data points to nearest centroid
4. Update centroids
5. Repeat until convergence

Q6. What are the applications of Clustering?

Applications:

- Customer segmentation
- Market analysis
- Image segmentation
- Recommendation systems
- Anomaly detection

Q7. What is the role of K value in K-Means?

K represents the number of clusters to be formed in the dataset.

BDA

Assignment 1:- Hadoop, BigQuery, Databricks, Snowflake, Amazon Redshift

Q1. What is Hadoop?

[Apache Hadoop](#) is an open-source framework used for distributed storage and processing of big data across multiple computers.

Q2. What is HDFS?

HDFS (Hadoop Distributed File System) is the storage system of Hadoop used to store large data across multiple nodes.

Q3. What is Google BigQuery?

[Google BigQuery](#) is a cloud-based data warehouse used for fast SQL queries and big data analytics.

Q4. What is Databricks Lakehouse Platform?

[Databricks](#) is a cloud platform that combines data warehousing and data lakes for analytics, machine learning, and big data processing.

Q5. What is Snowflake?

[Snowflake](#) is a cloud-based data warehousing platform used for storing, processing, and analyzing data.

Q6. What is Amazon Redshift?

[Amazon Redshift](#) is a cloud data warehouse service provided by AWS for large-scale data analytics.

Q7. What is the role of Cloudera Manager?

Cloudera Manager is used to:

- Set up Hadoop clusters
- Manage services
- Monitor cluster performance
- Configure Hadoop components

Q8. What is YARN in Hadoop?

YARN (Yet Another Resource Negotiator) is Hadoop's resource management system used to manage cluster resources and job scheduling.

Q9. What are the applications of Big Data platforms?

Applications:

- Data warehousing
- Real-time analytics
- Machine learning
- Business Intelligence
- Data visualization

Assignment 2:- MapReduce Word Count Program

Q1. What is MapReduce?

MapReduce is a programming model in Hadoop used for processing large data in parallel using two phases:

- Map phase

- Reduce phase

Q2. What is the purpose of the Mapper?

The Mapper:

- Reads input data
- Splits data into words
- Generates key-value pairs

Example:

(an,1)

(elephant,1)

Q3. What is the purpose of the Reducer?

The Reducer:

- Groups similar keys
- Adds values of same keys
- Produces final word count

Example:

(an,2)

Q4. What are the phases of MapReduce?

Main phases:

1. Map Phase
2. Shuffle and Sort Phase
3. Reduce Phase

Q5. What is Shuffle and Sort in MapReduce?

Shuffle and Sort groups all similar keys together before sending them to the Reducer.

Example:

All occurrences of “an” are grouped together.

Q6. What is a key-value pair in MapReduce?

A key-value pair is the basic data format used in MapReduce.

Example:

(word, count)

Example:

(cat,1)

Q7. What is Word Count Program in Hadoop?

Word Count is a MapReduce application used to count occurrences of words in a file.

Q8. What are the advantages of MapReduce?

Advantages:

- Parallel processing
- Fault tolerance
- Scalability
- Faster processing of big data

Assignment 3:- MapReduce Program to Find Student Grades

Q1. What is Shuffling and Sorting in MapReduce?

Shuffling and Sorting groups all intermediate key-value pairs with the same key before sending them to the Reducer.

Q2. What is the advantage of MapReduce in student grading?

Advantages:

- Parallel processing
- Faster computation
- Handles large student datasets
- Scalable and fault tolerant

Q3. What is the purpose of the Driver class?

The Driver class:

- Configures the MapReduce job
- Sets Mapper and Reducer classes
- Defines input and output paths
- Runs the job

Q4. What are the applications of this program?

Applications:

- Automated grading systems
- Educational data analysis

- Student performance evaluation

Q5. Why is MapReduce suitable for large datasets?

Because it:

- Processes data in parallel
- Distributes workload across multiple nodes
- Reduces processing time

Q6. What is scalability in MapReduce?

Scalability means the system can handle increasing amounts of data efficiently by adding more machines/nodes.

Q7. How can grading criteria be changed in this program?

Grading criteria can be changed by modifying the `calculateGrade()` method in the Reducer class.

Assignment 4:- MongoDB CRUD Operations

Q1. What is MongoDB?

MongoDB is a NoSQL, document-oriented database used to store data in JSON-like documents.

Q2. What is NoSQL Database?

A NoSQL database stores data in non-tabular format and is designed for handling large and unstructured data.

Q3. What does CRUD stand for?

CRUD stands for:

- Create
- Read
- Update
- Delete

These are the basic database operations.

Q4. How do you create a database in MongoDB?

Command:

```
use myNewDatabase
```

MongoDB creates the database when data is inserted for the first time.

Q5. What is a Collection in MongoDB?

A Collection is a group of MongoDB documents, similar to a table in relational databases.

Q6. How do you insert a document in MongoDB?

Command:

```
db.myNewCollection.insertOne({name:"John Doe", age:30})
```

Q7. How do you retrieve data from MongoDB?

Command:

```
db.myNewCollection.findOne({name:"John Doe"})
```

Used to query documents from a collection.

Q8. How do you update a document in MongoDB?

Command:

```
db.myNewCollection.updateOne(  
  {name:"John Doe"},  
  {$set:{age:31}}  
)
```

Q9. How do you delete a document in MongoDB?

Command:

```
db.myNewCollection.deleteOne({name:"Bob"})
```